

DQN and Double DQN for CartPole-v1

A Reproducible DATA8007 Reinforcement Learning Study

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DATA8007 Foundations of Sequential Decision-Making

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Project Question

- Can a compact PyTorch implementation reproduce DQN and Double DQN on Gymnasium CartPole-v1?
- Does Double DQN improve performance or stability under the same hyperparameters?
- How much seed sensitivity appears in a small deep RL experiment?

Problem Setting

- Environment: Gymnasium CartPole-v1.
- Observation: four continuous state variables.
- Action space: two discrete actions.
- Maximum episode length: 500 steps.
- Solved threshold: average return near 475.

- Q-network: MLP with hidden sizes [128, 128] and ReLU activations.
- Stabilization: experience replay and a target network.
- Optimization: Adam, Smooth L1 loss, batch size 64.
- Exploration: epsilon-greedy schedule from 1.0 to 0.05.

$$y^{DQN} = r + \gamma(1 - d) \max_{a'} Q_{\theta^-}(s', a')$$

Double DQN Comparison

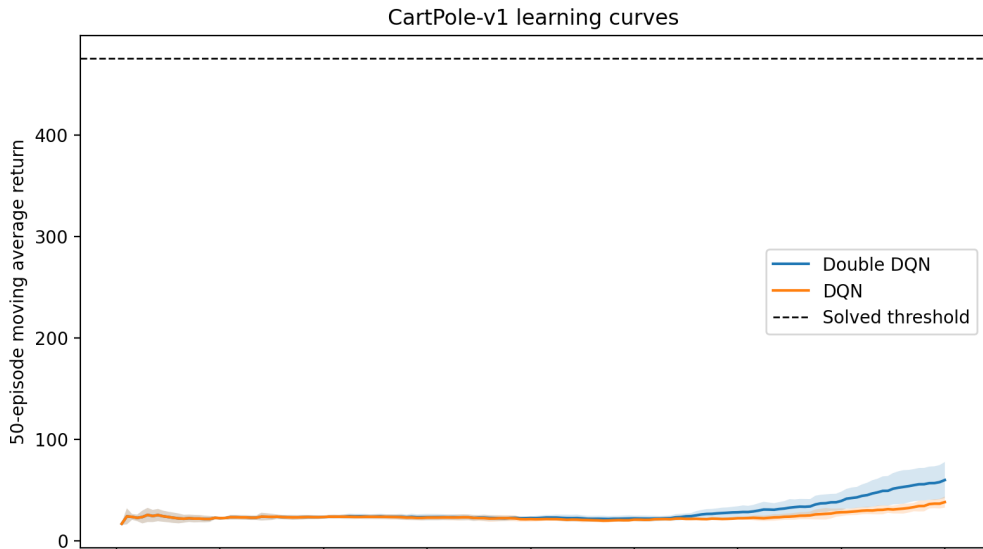
- Double DQN uses the online network to select the next action.
- The target network evaluates the selected action.
- This can reduce overestimation bias from the max operator.

$$a^* = \arg \max_{a'} Q_{\theta}(s', a'), \quad y^{Double} = r + \gamma(1 - d)Q_{\theta-}(s', a^*)$$

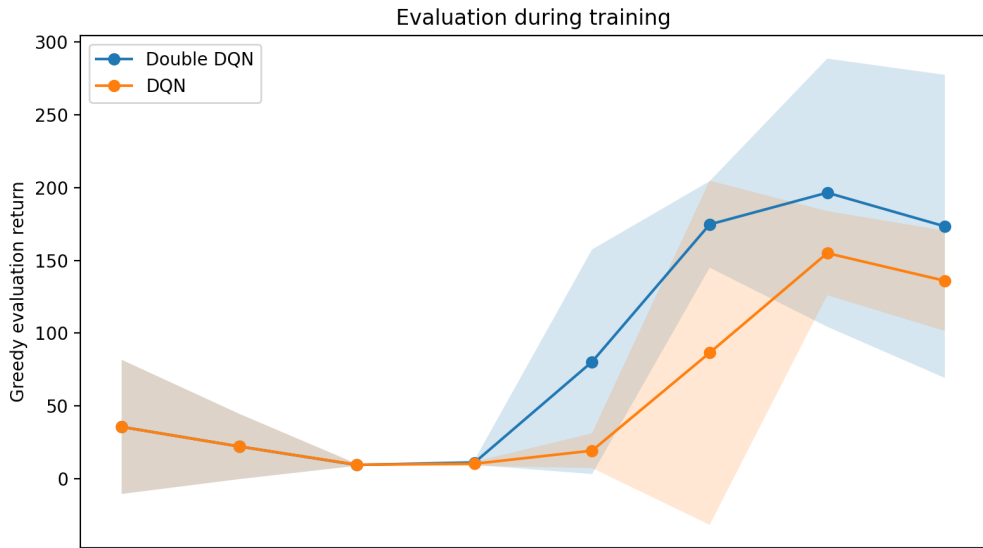
Experimental Design

- Algorithms: DQN and Double DQN.
- Seeds: 0, 1, and 2 for each method.
- Training budget: 160 episodes.
- Evaluation: five greedy episodes every 20 training episodes.
- Final checkpoint evaluation: 20 greedy episodes with evaluation seed 12345.
- Hardware: NVIDIA L40S GPU when CUDA is available.

Training Curves



Evaluation Curves



Summary Results

Method	Final return	Final MA50	Best eval
DQN	90.33 ± 37.45	38.37 ± 5.46	166.60 ± 48.91
Double DQN	160.33 ± 57.84	60.09 ± 17.75	250.93 ± 74.79

- Double DQN achieved higher aggregate returns under the fixed budget.
- No run reached the CartPole solved threshold.
- Variance across seeds remains substantial.

- Configuration files define method, seeds, device, and hyperparameters.
- Training, evaluation, and plotting scripts save logs and figures.
- Results are stored in CSV, JSON, and PNG files.
- The repository includes the LaTeX report and this slide deck PDF.

- The project implements a complete DQN and Double DQN workflow in PyTorch.
- Double DQN performed better in this controlled CartPole experiment.
- The conclusion is cautious because training was short and seed variance was high.
- Natural extensions: more seeds, longer training, prioritized replay, dueling networks, and harder environments.